

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style='whitegrid')
```

```
plt.style.use('seaborn-v0_8-darkgrid')
```

```
df = pd.read_csv('/content/Starbucks_stock_history.csv')
```

```
df['Date'] = pd.to_datetime(df['Date'], dayfirst=True)
df = df.sort_values('Date')
```

```
print("Shape:", df.shape)
print("Columns:", df.columns.tolist())
df.head()
```

```
↗ Shape: (7372, 8)
Columns: ['Date', 'Open', 'High', 'Low', 'Close', 'Volume', 'Dividends', 'Stock Splits']
```

	Date	Open	High	Low	Close	Volume	Dividends	Stock Splits
0	1992-06-26	0.270957	0.287086	0.264507	0.277409	224358400	0.0	0
1	1992-06-29	0.280635	0.303215	0.274183	0.296763	58732800	0.0	0
2	1992-06-30	0.303215	0.306440	0.283860	0.287086	34777600	0.0	0
3	1992-07-01	0.290312	0.296763	0.280635	0.293538	18316800	0.0	0
4	1992-07-02	0.296763	0.296763	0.287086	0.293538	13996800	0.0	0

```
print(df.isnull().sum())
```

```
↗ Date      0
  Open      0
  High      0
  Low       0
  Close     0
  Volume     0
  Dividends  0
  Stock Splits 0
  dtype: int64
```

```
Q1 = df['Volume'].quantile(0.25)
Q3 = df['Volume'].quantile(0.75)
IQR = Q3 - Q1
```

```
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR
df['Volume_capped'] = np.where(df['Volume'] > upper_bound, upper_bound,
                               np.where(df['Volume'] < lower_bound, lower_bound, df['Volume']))
```

```
print(df.describe())
print(df.dtypes)
df['Year'] = df['Date'].dt.year
df['Month'] = df['Date'].dt.month
print("Years in data:", df['Year'].unique())
```

```
↗
```

	Date	Open	High	Low	\
count	7372	7372.000000	7372.000000	7372.000000	
mean	2007-02-10 07:44:53.651654912	22.111116	22.328499	21.889586	
min	1992-06-26 00:00:00	0.270957	0.287086	0.264507	
25%	1999-10-12 18:00:00	3.002790	3.093206	2.922471	
50%	2007-02-12 12:00:00	9.756537	9.855629	9.634734	
75%	2014-06-09 06:00:00	33.884453	34.146608	33.633797	
max	2021-10-01 00:00:00	125.592534	125.831609	124.327440	
std	NaN	27.496121	27.725446	27.256726	

	Close	Volume	Dividends	Stock Splits	Volume_capped
count	7372.000000	7.372000e+03	7372.000000	7372.000000	7.372000e+03
mean	22.117324	1.531211e+07	0.001388	0.001628	1.436046e+07
min	0.277409	1.504000e+06	0.000000	0.000000	1.504000e+06
25%	2.999823	8.254150e+06	0.000000	0.000000	8.254150e+06
50%	9.763925	1.227860e+07	0.000000	0.000000	1.227860e+07
75%	33.889964	1.831200e+07	0.000000	0.000000	1.831200e+07
max	125.572609	5.855088e+08	0.450000	2.000000	3.339878e+07
std	27.500922	1.376366e+07	0.020384	0.057038	7.979972e+06
Date	datetime64[ns]				
Open	float64				
High	float64				

```

Low                float64
Close              float64
Volume             int64
Dividends          float64
Stock Splits       int64
Volume_capped      float64
dtype: object
Years in data: [1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005
2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019
2020 2021]

```

```

plt.figure(figsize=(12, 6))
plt.plot(df['Date'], df['Close'], label='Close Price', color='green')
plt.title('☕ Starbucks Closing Price Over Time')
plt.xlabel('Date')
plt.ylabel('Close Price')
plt.legend()
plt.show()

```

🔗 /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128200 (\N{CHART WITH UPWARDS TREND}) missing from font. Using fallback font.

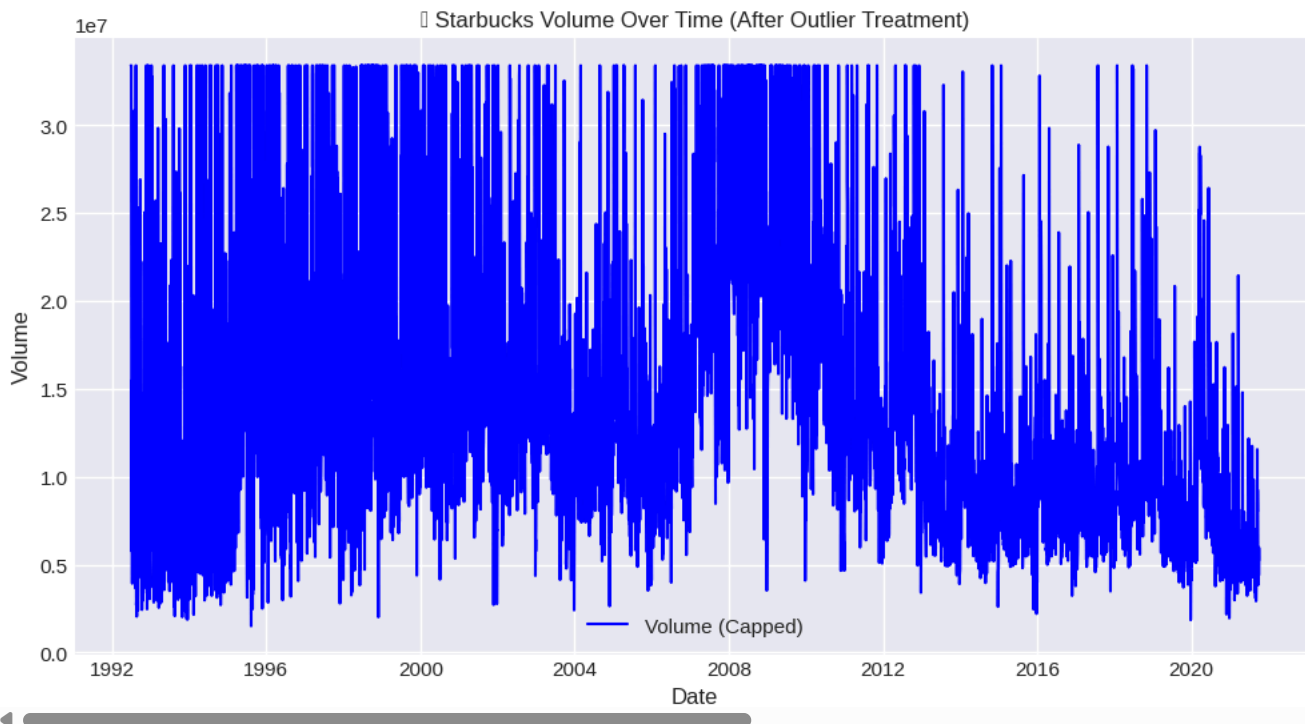


```

plt.figure(figsize=(12, 6))
plt.plot(df['Date'], df['Volume_capped'], label='Volume (Capped)', color='blue')
plt.title('📊 Starbucks Volume Over Time (After Outlier Treatment)')
plt.xlabel('Date')
plt.ylabel('Volume')
plt.legend()
plt.show()

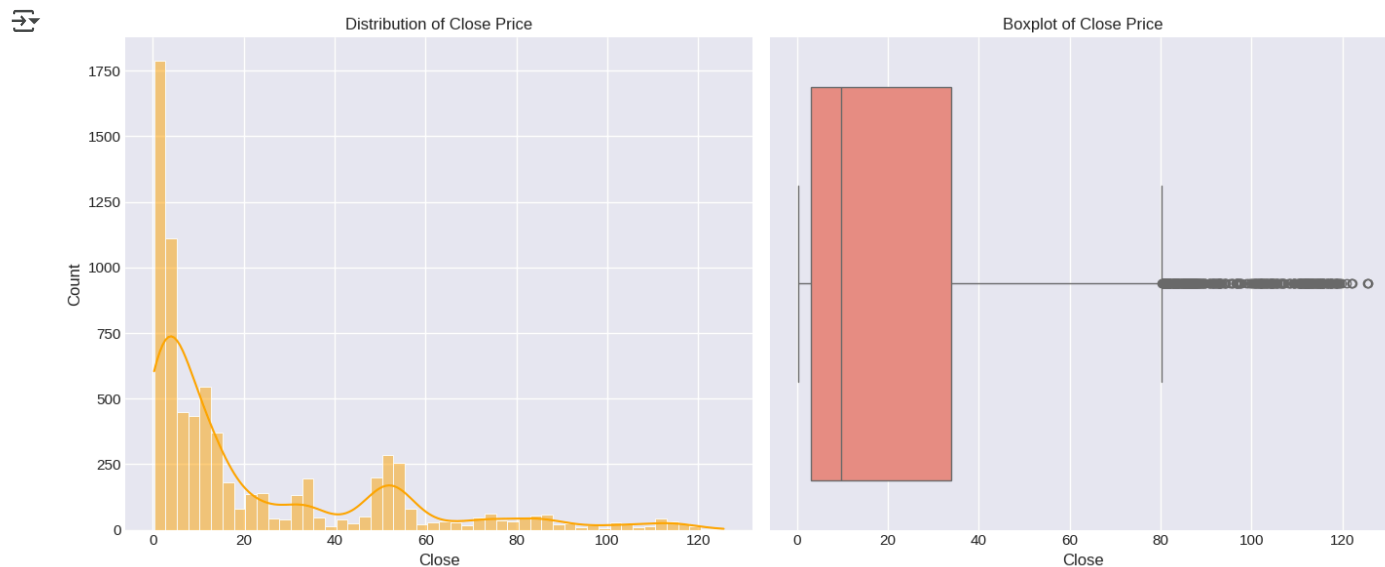
```

 /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128202 (\N{BAR CHART}) missing from font fig.canvas.print_figure(bytes_io, **kw)



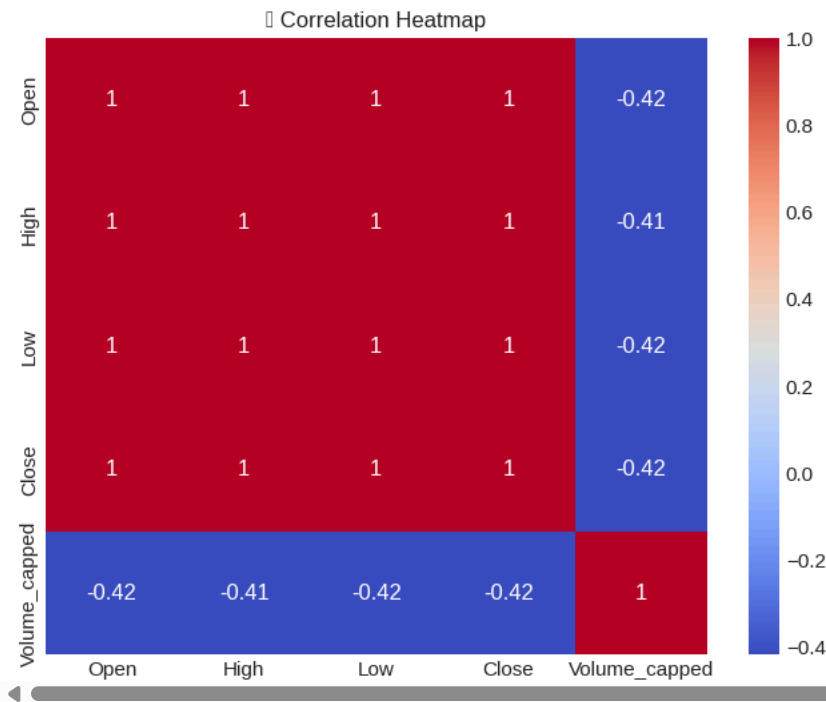
```
plt.figure(figsize=(14, 6))
plt.subplot(1, 2, 1)
sns.histplot(df['Close'], bins=50, kde=True, color='orange')
plt.title('Distribution of Close Price')
plt.subplot(1, 2, 2)
sns.boxplot(x=df['Close'], color='salmon')
plt.title('Boxplot of Close Price')
```

```
plt.tight_layout()
plt.show()
```



```
plt.figure(figsize=(8, 6))
sns.heatmap(df[['Open', 'High', 'Low', 'Close', 'Volume_capped']].corr(), annot=True, cmap='coolwarm')
plt.title("🔥 Correlation Heatmap")
plt.show()
```

 /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128293 (\N{FIRE}) missing from font(s) Li
fig.canvas.print_figure(bytes_io, **kw)



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib
sns.set(style='whitegrid')
plt.style.use('seaborn-v0_8-darkgrid')
```

```
!apt install -y fonts-dejavu
```

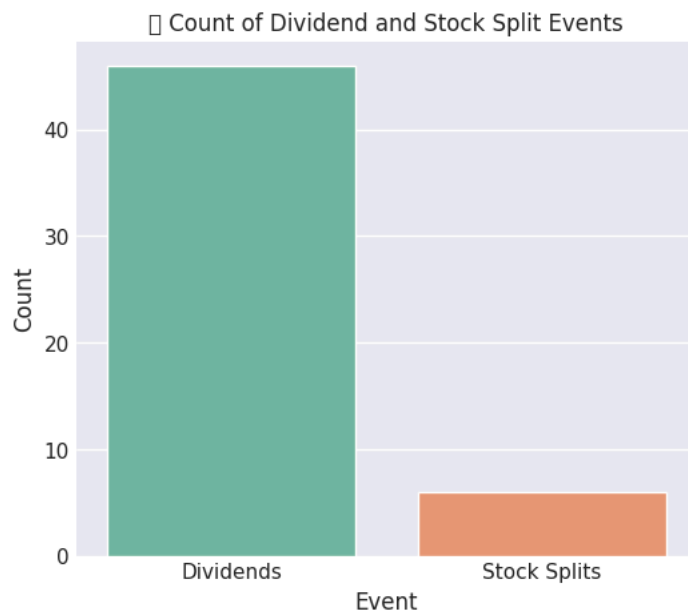
```
df['Dividends_Flag'] = df['Dividends'] > 0
df['Splits_Flag'] = df['Stock Splits'] > 0
event_counts = pd.DataFrame({
    'Dividends': [df['Dividends_Flag'].sum()],
    'Stock Splits': [df['Splits_Flag'].sum()]
}).melt(var_name='Event', value_name='Count')
```

```
plt.figure(figsize=(6, 5))
matplotlib.rcParams['font.family'] = 'DejaVu Sans'
sns.barplot(x='Event', y='Count', data=event_counts, palette='Set2', hue='Event', legend=False) # Added hue='Event' and legend=False
plt.title('🔥 Count of Dividend and Stock Split Events')
plt.show()
```

```

↳ Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  fonts-dejavu-core fonts-dejavu-extra
The following NEW packages will be installed:
  fonts-dejavu fonts-dejavu-core fonts-dejavu-extra
0 upgraded, 3 newly installed, 0 to remove and 30 not upgraded.
Need to get 3,085 kB of archives.
After this operation, 10.7 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-dejavu-core all 2.37-2build1 [1,041 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-dejavu-extra all 2.37-2build1 [2,041 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/universe amd64 fonts-dejavu all 2.37-2build1 [3,192 B]
Fetched 3,085 kB in 0s (18.2 MB/s)
Selecting previously unselected package fonts-dejavu-core.
(Reading database ... 126315 files and directories currently installed.)
Preparing to unpack .../fonts-dejavu-core_2.37-2build1_all.deb ...
Unpacking fonts-dejavu-core (2.37-2build1) ...
Selecting previously unselected package fonts-dejavu-extra.
Preparing to unpack .../fonts-dejavu-extra_2.37-2build1_all.deb ...
Unpacking fonts-dejavu-extra (2.37-2build1) ...
Selecting previously unselected package fonts-dejavu.
Preparing to unpack .../fonts-dejavu_2.37-2build1_all.deb ...
Unpacking fonts-dejavu (2.37-2build1) ...
Setting up fonts-dejavu-core (2.37-2build1) ...
Setting up fonts-dejavu-extra (2.37-2build1) ...
Setting up fonts-dejavu (2.37-2build1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 127974 (\N{BANK}) missing from font(s)
fig.canvas.print_figure(bytes_io, **kw)

```



```

df['Month_Name'] = df['Date'].dt.strftime('%b')

plt.figure(figsize=(12, 6))
sns.boxplot(x='Month_Name', y='Close', data=df,
            order=['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
                  'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec'])
plt.title("Monthly Seasonality in Closing Price")
plt.xlabel("Month")
plt.ylabel("Close Price")
plt.show()

```

```

/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128198 (\N{TEAR-OFF CALENDAR}) missing from font
fig.canvas.print_figure(bytes_io, **kw)

```



```

df['MA_30'] = df['Close'].rolling(window=30).mean()
df['MA_100'] = df['Close'].rolling(window=100).mean()

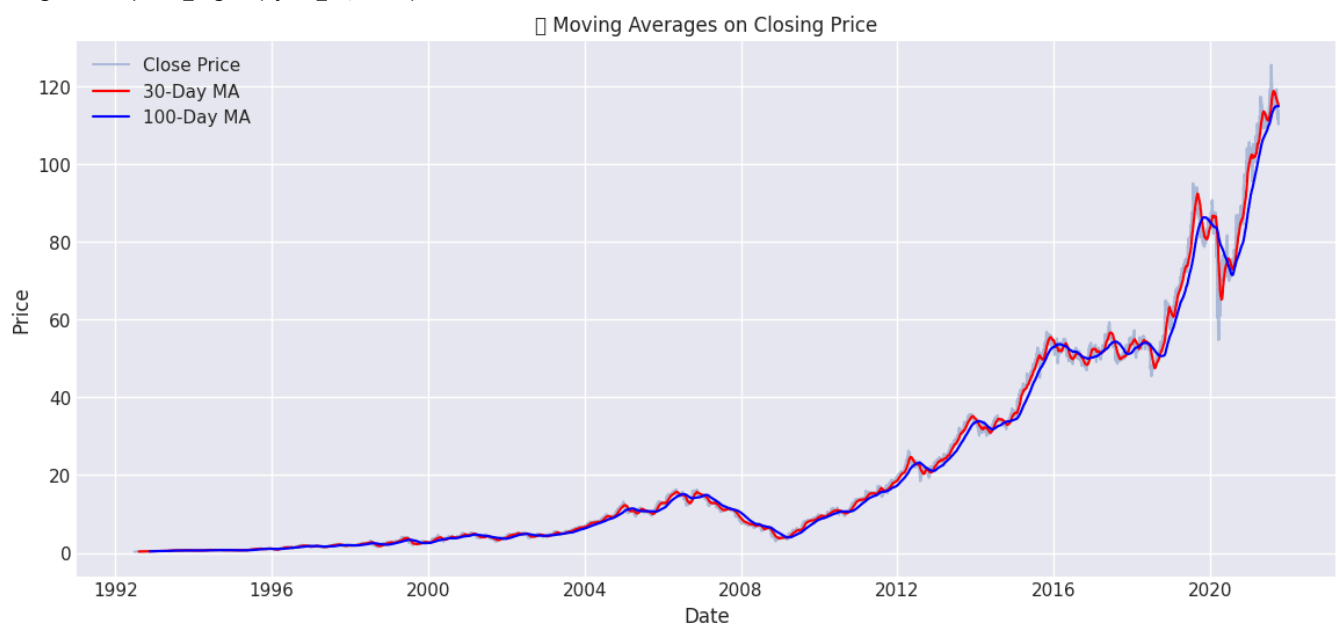
plt.figure(figsize=(14, 6))
plt.plot(df['Date'], df['Close'], label='Close Price', alpha=0.4)
plt.plot(df['Date'], df['MA_30'], label='30-Day MA', color='red')
plt.plot(df['Date'], df['MA_100'], label='100-Day MA', color='blue')
plt.title('Moving Averages on Closing Price')
plt.xlabel('Date')
plt.ylabel('Price')
plt.legend()
plt.show()

```

```

/usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128202 (\N{BAR CHART}) missing from font
fig.canvas.print_figure(bytes_io, **kw)

```



```

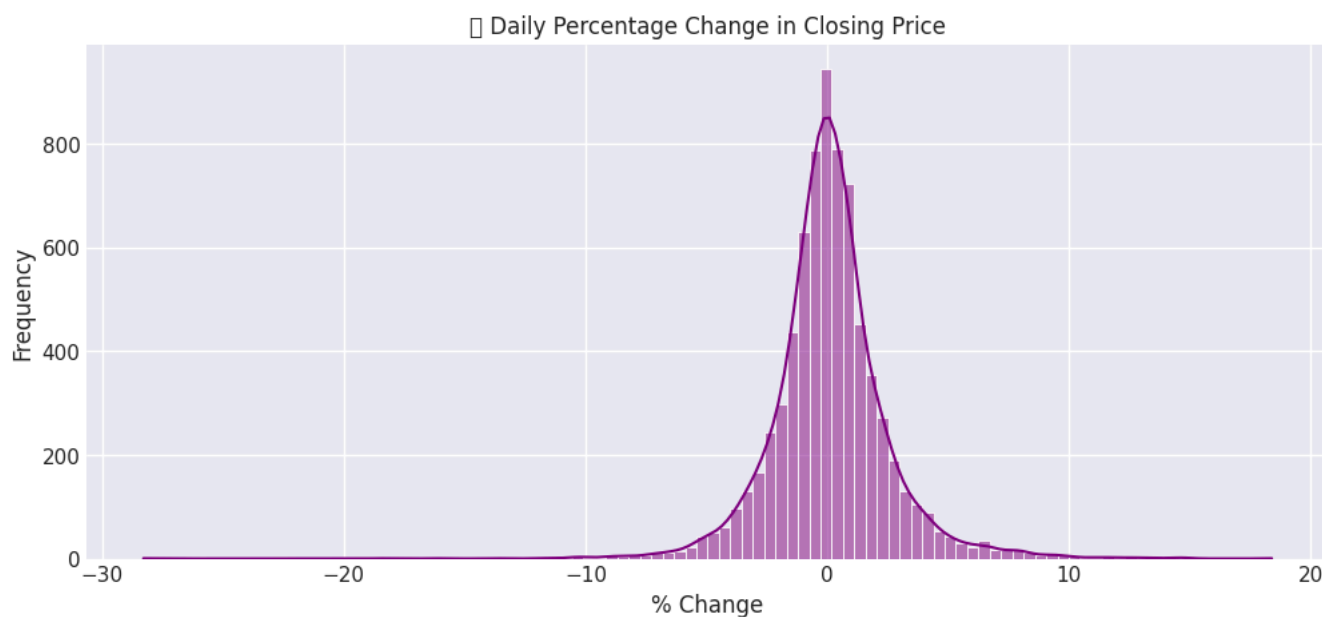
df['Daily Change %'] = df['Close'].pct_change() * 100

plt.figure(figsize=(12, 5))
sns.histplot(df['Daily Change %'].dropna(), bins=100, color='purple', kde=True)

```

```
plt.title('📉 Daily Percentage Change in Closing Price')
plt.xlabel('% Change')
plt.ylabel('Frequency')
plt.show()
```

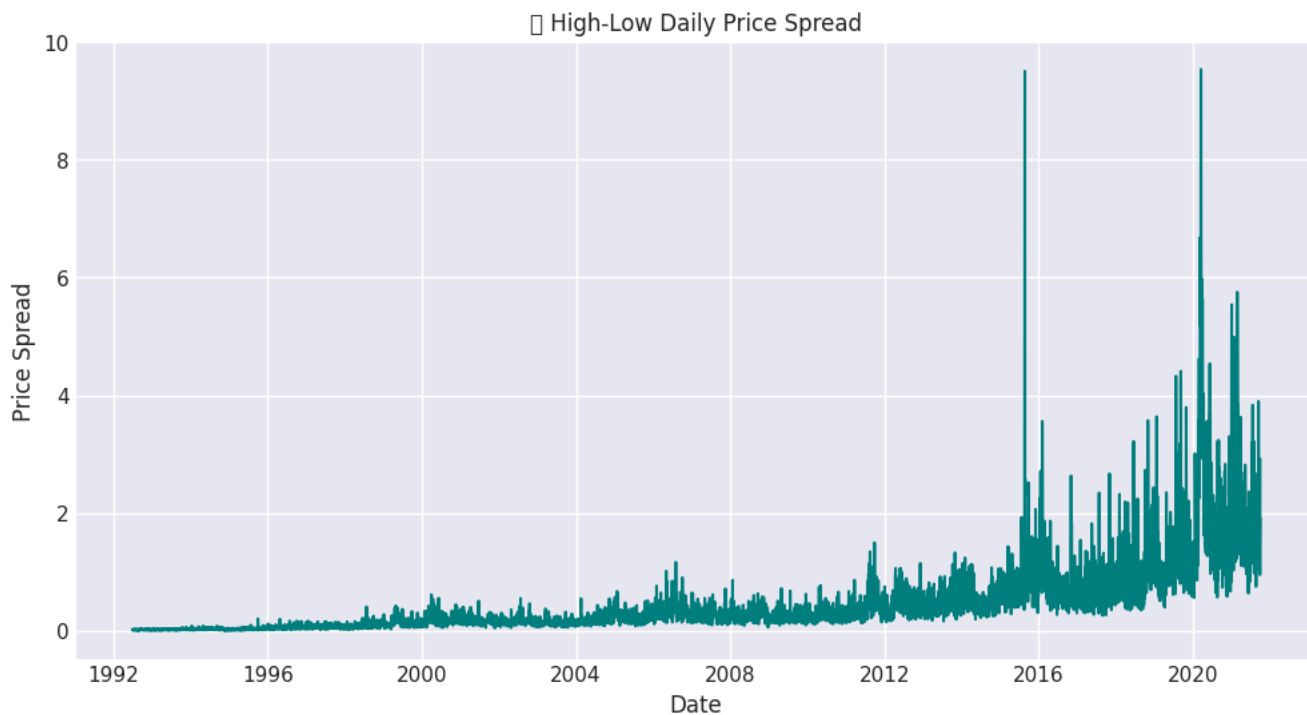
↗ /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128201 (\N{CHART WITH DOWNWARDS TREND}) not found. fig.canvas.print_figure(bytes_io, **kw)



```
df['HL Spread'] = df['High'] - df['Low']
```

```
plt.figure(figsize=(12, 6))
plt.plot(df['Date'], df['HL Spread'], color='teal')
plt.title('📈 High-Low Daily Price Spread')
plt.xlabel('Date')
plt.ylabel('Price Spread')
plt.show()
```

↗ /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128260 (\N{ANTICLOCKWISE DOWNWARDS AND UPWARDS TRIANGLE}) not found. fig.canvas.print_figure(bytes_io, **kw)



```
yearly_avg = df.groupby(df['Year'])['Close'].mean().reset_index()
```

```
plt.figure(figsize=(10, 5))
sns.lineplot(data=yearly_avg, x='Year', y='Close', marker='o', color='darkblue')
plt.title('📊 Yearly Average Close Price')
plt.xlabel('Year')
```

```
plt.ylabel('Avg Close Price')  
plt.grid(True)  
plt.show()
```

 /usr/local/lib/python3.11/dist-packages/IPython/core/pylabtools.py:151: UserWarning: Glyph 128197 (\N{CALENDAR}) missing from font(s)
fig.canvas.print_figure(bytes_io, **kw)

